Lab 3: Facial Expression Recognition

Sophia Thurston

11/10/2026

How well does the baseline work?

The baseline fine-tunes an ImageNet-pretrained ResNet-18 on FER2013 with the 8-class FER+ labels, enlarging each 48×48 face to 128×128. After 25 epochs it reached 85.5% test accuracy and 0.723 macro F1, with the best validation checkpoint at epoch 21. The common classes were recognized well, with F1 of 0.949 for happy and 0.867 for neutral, but the rare ones were not, with 0.480 for contempt and 0.450 for disgust. These two classes have under 180 training images each, against 9,365 for neutral, and only 26 and 22 test images, so a change of two or three test images moves their F1 by about 0.06 to 0.11. Because each model below was trained only once, we treat differences smaller than about one point of accuracy or 0.02 macro F1 as possibly due to noise. Therefore, the baseline is strong on common expressions, and most of the room for improvement is in the rare classes, which macro F1 captures much better than accuracy.

Should the network be adapted to small images?

ResNet-18 begins with a 7×7 stride-2 convolution and a max-pool, which shrink the input by 4× before the first residual block. Feeding it the 48×48 faces directly, without enlarging them, lowered the results to 81.8% accuracy and 0.683 macro F1, because the face is only 12×12 when the residual blocks start and 2×2 at the end. We then adapted the network by replacing the first convolution with a 3×3 stride-1 convolution, initialized from shrunken copies of the pretrained filters, and removing the max-pool. All other layers kept their ImageNet weights. The adapted network reached 86.1% accuracy and 0.738 macro F1, 0.6 and 0.015 above the enlarged baseline, with disgust improving from 0.450 to 0.545. Its final feature map is 6×6 rather than the baseline's 4×4, so more spatial detail around the eyes and mouth is kept, while enlarging the image only adds interpolated pixels with no new information. The cost was speed, as each epoch took 24 s instead of 14 s. Therefore, working at the original resolution is better than enlarging the faces, but only if the network is adapted, since skipping the enlargement alone costs about four points.

Can training be improved?

The original FER+ paper (Barsoum et al.) reported 83.85% with a VGG-13 trained on majority-vote labels and 84.99% with its best label scheme, so the baseline already matches it. More recent methods report 88.55% (RAN), 89.35% (SCN), and 90.83% (TransFER). These numbers are not exactly comparable because our code drops test images whose top annotator votes were tied or went to "unknown" or "not a face".

Training for 80 epochs instead of 25 gave 86.2% and 0.736 macro F1. Adding an exponential moving average (EMA) of the weights gave the same 86.2% and 0.737, though it made validation

much smoother, reaching 0.763 validation F1 by epoch 19 where the plain run needed until epoch 52. EMA added little at the end because the one-cycle schedule already decays the learning rate to almost zero, which settles the weights in a similar way. Adding mixup and CutMix on top lowered macro F1 to 0.716, below even the baseline, at almost unchanged accuracy (86.0%). The common classes improved slightly, but disgust fell from 0.533 to 0.409 and fear from 0.667 to 0.624. Mixing pairs most rare faces with common ones, and CutMix sets its label by pasted area even though the expression is concentrated in the small mouth and eye regions.

Our proposed method combines the parts that helped: the adapted 48×48 network from the last exercise, 80 epochs and EMA, without mixup. This reached 85.9% accuracy and 0.744 macro F1, which was the best macro F1 of any training change and 0.021 above the baseline. The gain came mostly from the rare classes, with contempt rising from 0.480 to 0.533 and fear from 0.621 to 0.667.

However, its accuracy was only 0.4 points above the baseline and slightly below the plain 80-epoch run. The checkpoint was chosen by validation macro F1 at epoch 65, while validation accuracy had peaked between epochs 42 and 50, so selecting for F1 traded away a little accuracy. Therefore, combining original resolution, longer training, and EMA improves macro F1 by about two points, but no training change improved accuracy by a full point, and mixup hurt the rare classes it was expected to help.

How much does pretraining matter?

Training ResNet-18 from scratch reached 82.5% accuracy and 0.678 macro F1, 3.0 points and 0.045 below the ImageNet-pretrained baseline. Its training was also unstable early on, as validation accuracy fell to 41.9% at epoch 6, while the pretrained model never dropped below 75% after the first epoch. AlexNet, the network used in the AffectNet paper, reached only 81.7% and 0.683 after 30 epochs, below the pretrained ResNet-18 and level with ResNet-18 trained from scratch, despite having 57.0M parameters against 11.2M.

Comparing the training curves, ResNet-18 learned faster and further. It reached 84.3% validation accuracy by epoch 12, while AlexNet was still at 80.0% by epoch 14 and ended at 82.5%. At the end, ResNet-18's train and validation losses were 1.047 and 1.281, against 1.265 and 1.344 for AlexNet, and its best validation macro F1 was 0.764 against 0.703. Neither loss can approach zero, because of label smoothing and the class weights. The gap between train and validation loss tells the models apart. For ResNet-18, validation loss barely improved after about epoch 13 while training loss kept falling, a gap of 0.23 that shows mild overfitting, though validation scores did not suffer. AlexNet's gap was only 0.08, so it was underfitting rather than overfitting, as its dropout and our augmentation keep its large fully connected layers in check. ResNet-18 does better with fewer parameters because its residual connections and batch normalization let a deeper network train well, while most of AlexNet's parameters sit in fully connected layers that add capacity but not better visual features.

Lastly, the network used was InceptionResNet-V1 pretrained to recognize the 8,631 people in the VGGFace2 training set (over 3 million photos), fine-tuned at 160×160 with the pretrained layers at a tenth of the classifier's learning rate. It was chosen because, unlike ImageNet, its training data is entirely faces under varied pose, lighting, and age. It reached 86.8% accuracy and 0.747 macro F1, the best of any model, 1.3 points and 0.024 above the baseline, 4.3 points above training from scratch, and 5.1 points above AlexNet. It was also useful immediately, with 81.0% validation accuracy after one epoch, against 67.6% for the ImageNet baseline and 42.4% from scratch. This is notable because recognition training teaches a network to give the same output for a person

whatever their expression. Its early and middle layers still learn where facial features are and how faces vary, and fine-tuning every layer let it recover the expression information its final layers had learned to ignore. However, this comparison also changes the architecture (23.5M parameters) and the input size, so part of the gain may come from those rather than the pretraining. Therefore, pretraining matters and pretraining on faces helps more than pretraining on everyday objects, even when the original task was identity rather than expression.

How does the model do on new photos?

I took six photos of myself (angry, disgusted, happy, neutral, sad, and surprised) and classified them with the best model, the face-pretrained InceptionResNet-V1. Five of the six were predicted correctly: disgusted (92.8%), surprised (78.2%), neutral (77.2%), happy (76.6%), and sad (54.1%). The angry photo was predicted as disgust (38.0%), with neutral second (29.6%) and anger only third (18.8%). Disgust and anger were also often confused by both the annotators and the models in the AffectNet paper. Sad was the least confident correct prediction, with neutral second at 25.7%, which fits sad being one of the weaker classes on the test set (F1 0.760). My photos also differ from FER2013, as they are deliberately posed, frontal, and from a single person, and I did not include fear or contempt, so six photos illustrate the model's behavior rather than measure its accuracy. Therefore, the model generalizes reasonably to new faces once they are cropped the same way as the training data, its one mistake follows a confusion also seen in the AffectNet paper, and matching the training preprocessing matters as much as the model itself.